

Question Number	Answer	Additional Guidance	Mark
1(a)(i)	• {diameter / area} of zone of inhibition/ number of colonies (on agar) / degree of turbidity / transmission/ absorbance / number of cells (1) • mm / mm²/ number (cells) per cm³ / % (transmission) / au (absorbance) (1)	Accept width Accept suitable alternative units (e.g. cm³)	2

Question Number	Answer	Additional Guidance	Mark
1(a)(ii)	An answer that includes the following points: • description of how bacteria are cultured (1) • description of how extracts are added to culture (1) • control of feature of independent variable described (1) • incubate at suitable temperature for suitable time (1) • appropriate measurement of DV (1) • ref to relevant safety measure (1)	lawn /pour plate / broth e.g. put filter paper discs soaked on culture / cut wells e.g disc size /extract volume / solvent volume temp. <37, time 1 day to 7 days e.g. disinfect equipment, flame equipment, lift lid partially, autoclaving after, goggles, gloves, tie back long hair	5

Question Number	Answer	Additional Guidance	Mark
1(a)(iii)	An answer that includes the following steps: • correct reading of figures from graph (1) • calculation of percentage change (1)	29, 14 52 (%) OR 110 (%)	2

Question Number	Answer	Additional Guidance	Mark
1(a)(iv)	An answer that includes the following points: • All the extracts have an effect on all of the bacteria (1) • methanol extract has greatest effect (on all bacteria) (1) • both extracts are more effective than gentamycin (in all four bacteria) / acetone extract less effective than methanol extract (1) • greatest effect is on {<i>S aureus</i> / <i>S typhimurium</i>} / least effect is on <i>P. aeruginosa</i> / (1)		(4)

Question Number	Answer	Additional Guidance	Mark
1(b)	An answer including the following points : • {all 3 / both} inhibit bacterial growth in all (4) bacteria (1) • methanol extract more effective than both chemicals (1) • (in each case) the effect of the methanol extract is more than the sum of the effect of the two chemicals (1) • there may be other anti-microbial chemicals in the methanol extract / the two chemicals work together to produce an effects greater than their sum (synergy) (1)		(4)

Question Number	Answer	Additional Guidance	Mark
1(c)	An answer including three of the following: Placebos • it is used as a control / more validity (1) • to {eliminate} / account for} a psychological effect (of taking the drug) (1) Double-blind trials • so that {doctor / scientist} does not know if the treatment contains the (new) drug or not (1) • to remove bias from results (1)	e.g. comparison with the actual drug Accept to {eliminate} / account for} a psychological effect (of administering the drug)	(3)